Student Performance Prediction & Early Intervention

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Project Overview

Goal

Predict student **pass/fail** outcomes using machine learning to enable timely academic intervention.

Models Used

- Logistic Regression
- Decision Tree
- Gradient Boosting

Dataset

UCI Student Performance Dataset
395 records $\times$ 34 features
(student-mat.csv)

Key Technologies

Demographics · Attendance · Study time
LIME explainability
Streamlit web deployment

Problem Statement

Late Identification

At-risk students are identified **only after failing** — too late for effective support.

No Proactivity

Without predictive tools, teachers **cannot act proactively** on early warning signs.

Academic Impact

Delayed support **reduces student retention** and academic performance outcomes.

We need a **data-driven system** that predicts student outcomes *early* and enables **timely intervention** before it is too late.

Objectives

[1] **Predict** student pass/fail outcomes using ML models trained on academic, demographic, and behavioral data.

[2] **Identify** key factors influencing performance — study habits, attendance, prior grades — through thorough exploratory analysis.

[3] **Enable** actionable early interventions by providing counselors and teachers with transparent, explainable predictions.

Dataset Overview

395 Records

34 Features

3 Categories

Academic Features

- Study time
- Previous failures
- Extra classes
- School support

Demographic Features

- Age & Gender
- Parent education
- Family size
- Travel time

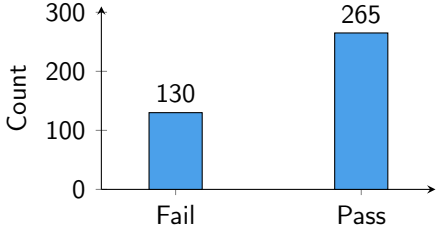
Behavioral Features

- Absences
- Social activities
- Health status
- Alcohol consumption

Source: UCI Machine Learning Repository — *student-mat.csv*

Exploratory Data Analysis — Key Findings

Class Distribution



265 pass vs 130 fail — class imbalance noted

Study Time vs Result

Students with higher study time **significantly outperform** peers.

Absences vs Result

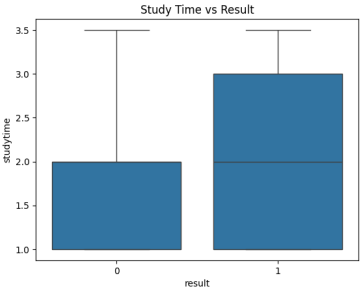
Higher absences **strongly correlate** with failure. Wider IQR for failing students.

Strongest Predictor

Past academic performance is the single strongest predictor of final outcomes.

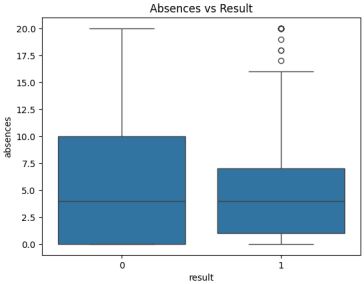
EDA — Visual Analysis

Study Time vs Result



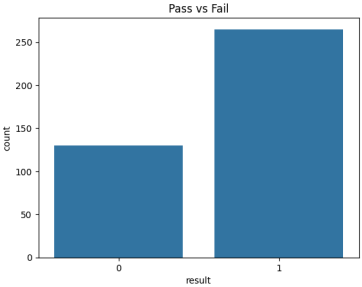
Higher study time → higher pass rate

Absences vs Result



More absences → higher failure risk

Pass vs Fail Distribution



≈2:1 pass/fail ratio in dataset

All plots from UCI *student-mat.csv* — IQR outlier capping applied before visualization

Data Preprocessing

Outlier Handling

- Outliers in `studytime` & `absences`
- IQR method to cap extreme values
- All major features clean post-treatment
- **Improved data quality & reliability**

Feature Engineering

- Categorical encoding via `pd.get_dummies`
- Feature selection: retained key predictors
- Target: `result` (pass=1 / fail=0)
- Train/test split for unbiased evaluation

Raw Data → Outlier Removal → Encoding → Feature Selection → Model Input

Machine Learning Models

Logistic Regression

- Baseline binary classifier
- Interpretable coefficients
- Fast training
- Robust on small datasets

Decision Tree

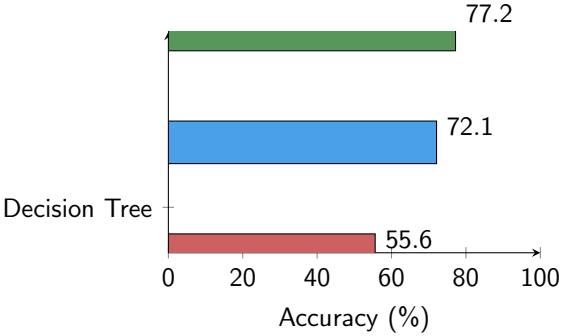
- Rule-based thresholds
- Highly interpretable
- Handles non-linearity
- No feature scaling needed

Gradient Boosting

- Ensemble of weak learners
- High predictive power
- Mixed feature types
- Captures complex patterns

All three models were trained and evaluated —
the best performer was selected for deployment

Model Performance Results



Best — Logistic Regression

77.2% accuracy

Selected for deployment

Runner-Up — Gradient Boosting

Accuracy: **72.1%**

Strong on complex patterns

Baseline — Decision Tree

Accuracy: **55.6%**

Interpretable but lower accuracy

Model Evaluation

Pass Prediction

Model identifies passing students with **high precision** — reliable for confirming likely success.

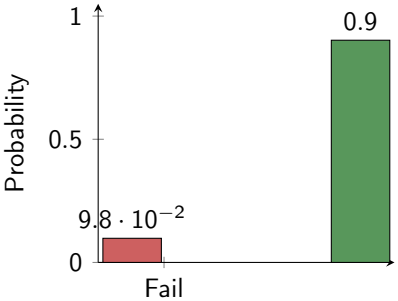
Fail Detection

Failing students are harder to detect — **lower recall** requires additional monitoring strategies.

Class Imbalance

More passing students than failing in dataset, affecting model **sensitivity to at-risk cases**.

Prediction Probabilities (Sample)

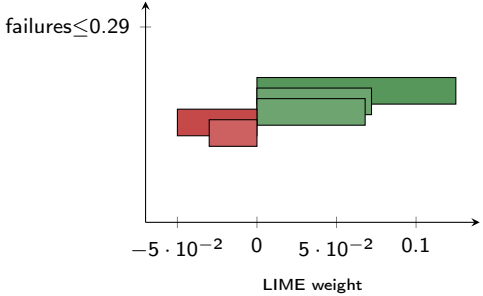


Example: 90.2% confidence — **Pass**

LIME — Model Explainability

How LIME Works

1. Perturbs input data around a student's record
2. Trains a simple linear model locally
3. Outputs which features pushed toward Pass or Fail



Key Capabilities

Per-Student Individual explanation for each prediction

Counselor-Ready Directly guides teacher interventions

Model-Agnostic Works with any ML model

Factors pushing toward Pass:

- ▲ Few past failures
- ▲ High study time
- ▲ Mother's education level

Factors reducing pass probability:

- ▼ Mother in services sector
- ▼ Older student age

Deployment — Streamlit Application

Application Workflow

1. Input
Teacher enters student details

2. Predict
Model outputs Pass/Fail instantly

3. Explain
LIME shows key driving factors

4. Act
Counselor plans intervention

Interface Highlights

Student Input Form

Sliders for age, study time, absences, failures & parent education; dropdowns for categorical fields

Prediction Output

Clear Pass/Fail result with confidence percentage and probability bar chart

LIME Visualization

Color-coded bar chart: green bars push toward Pass, red toward Fail

GitHub Repository & Collaboration

Version Control

All project code tracked with commits and branches — complete traceability of every change.

Collaboration

Branch-based workflow enabled Keerthana, Krishnendhu, and Theertha to work simultaneously without conflicts.

Project History

Full commit history documents progress from initial setup to final deployment.

GitHub Repository

`https://github.com/Keerthanajp18/Student-Performance-Prediction-and-Early-Intervention`

Limitations & Future Work

Limitations

- Small dataset limits model generalization
- Class imbalance reduces accuracy for fail prediction
- Limited features miss real-world behavioral factors

Future Work

- Use larger, diverse datasets for better accuracy
- Apply advanced models and imbalance handling (SMOTE)
- Integrate real-time data with personalized recommendations

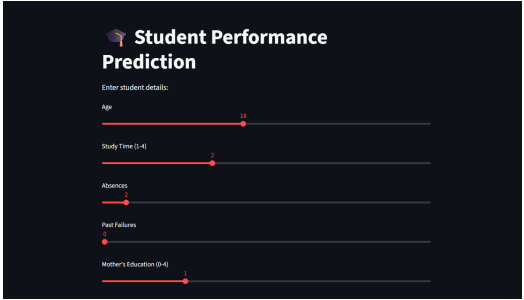
Addressing these limitations will significantly enhance model performance, scalability, and real-world impact in educational systems.

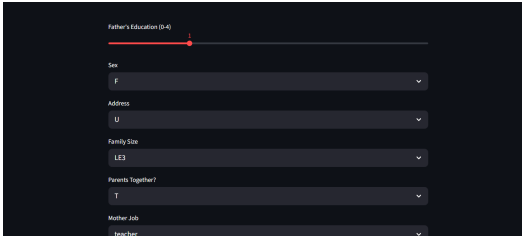
Conclusion

- ✓ ML **effectively predicts** student pass/fail outcomes
- ✓ Logistic Regression achieved **77.2% accuracy** — best model
- ✓ **LIME explainability** builds trust and enables actionable interventions
- ✓ **Streamlit deployment** makes the system accessible in real educational environments
- ✓ System is **scalable** and adaptable to any student dataset

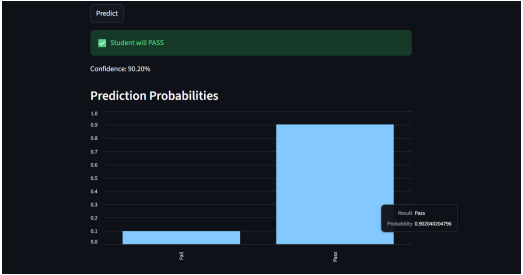
Deployed Application — Screenshots

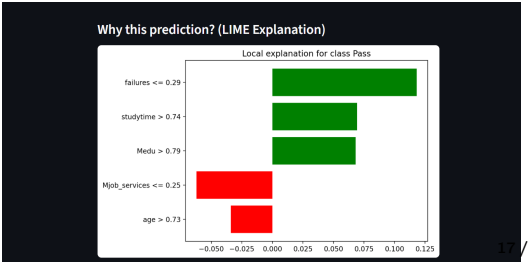
Input Screens





Output Screens





Thank You!

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